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Analysis of Information Technology in Boarding House

Group 10

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PROJECT ON

Analysis of Information Technology in Boarding House

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Analysis of Information Technology in Boarding House

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BACKGROUND

Along with the rapid development of information technology and the increasing complexity of human needs, the role of technology has become essential in solving daily problems. In the context of a boarding house (kost), information technology is required to optimize management, communication, and service delivery between tenants and the management. To support this, telecommunication technology through computer networks plays an important role in connecting devices, systems, and users.

The integration of information and telecommunication technology significantly affects how systems work, shifting from manual processes to digital and automated systems. This transition allows multiple devices or computers to communicate and collaborate in handling various tasks efficiently.

Currently, many boarding house management systems still rely on manual methods, which often lead to inefficiencies. These include errors in tenant data recording, delays in payment confirmation, difficulties in monitoring utilities usage, and miscommunication between management and tenants. Such issues reduce work efficiency and can negatively impact tenant satisfaction.

Therefore, the implementation of information technology in kost management is crucial. By applying digital systems and networks, managers can simplify administrative tasks, reduce human errors, and provide faster services to tenants. This project will discuss the application of information technology in boarding houses, including the required applications, hardware, operating systems, network infrastructure, benefits for management and tenants, as well as the estimated costs for implementation.

SYSTEM ANALYSIS

In designing the IT system for the Boarding House, a system analysis is conducted to understand the needs and existing problems in order to create an appropriate technological solution. The analysis reveals that the required system should include features such as digital room booking and registration, integrated payment systems, tenant management, automated financial and utility reporting and a digital complaint/request system. The system must also meet non-functional requirements such as data security, user-friendliness, high availability, and responsiveness.

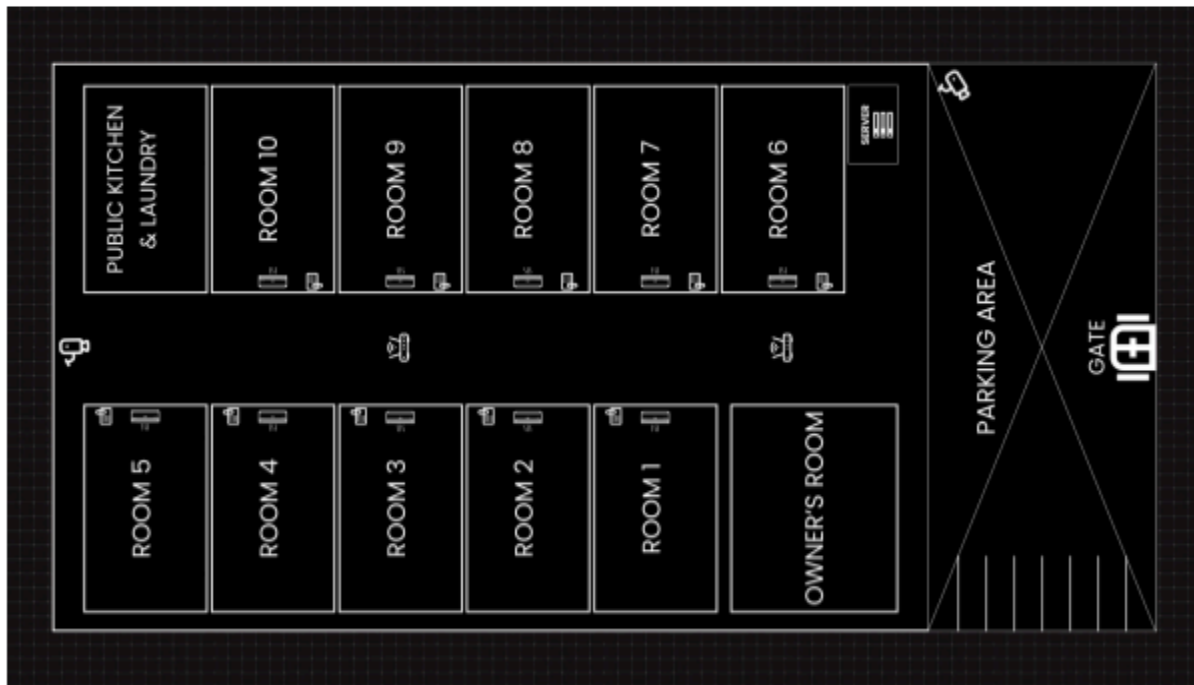
The main issues identified include manual booking and payment processes that cause inefficiencies, difficulties in monitoring tenant data, decentralized management of rent and utilities, and the absence of automated reporting and service request systems. These problems can be addressed through the implementation of an integrated IT system.

A SWOT analysis shows that the system offers strengths in improving efficiency, transparency, and tenant satisfaction, although challenges such as initial implementation costs and dependence on internet-connected devices remain.

Key stakeholders in this project include the boarding house owner, tenants, admin/management staff, IT team, and maintenance staff.

A feasibility study indicates that the implementation of this system is viable in terms of technical, economic, operational, and time aspects, with an estimated completion time of three to six months. With an integrated system in place, the boarding house operations can run more efficiently, modernize, and stay competitive in the digital era.

DESIGN BUILDING



A boarding house (kosan) is a residential facility that provides affordable and practical living spaces, usually designed for students or workers. In the context of modern management, a kosan can integrate digital solutions such as online booking, payment systems, and tenant management applications to improve efficiency and tenant comfort.

Layout:

- **Rooms:** The kosan consists of 10 private rooms, each already equipped with air conditioning to ensure comfort for tenants. The rooms are arranged neatly along both sides of the building.
- **Parking Area:** A dedicated parking space is located at the front, secured with CCTV and supported by WiFi internet coverage.
- **Connectivity:** The kosan is covered by a WiFi network that reaches all rooms, ensuring reliable internet access.
- **Management System:** The kosan uses a digital management application that handles tenant registration, payments, and facility monitoring.

Conclusion:

This layout provides a functional boarding house design with essential facilities such as AC-equipped rooms, laundry, secure parking, and internet connectivity, all supported by digital management for a modern and convenient living environment.

SOFTWARE

Number	Picture	Software
1		WhatsApp Business
2		BCA Mobile
3		Superkos
4		iPadOS 16

HARDWARE SPECIFICATION

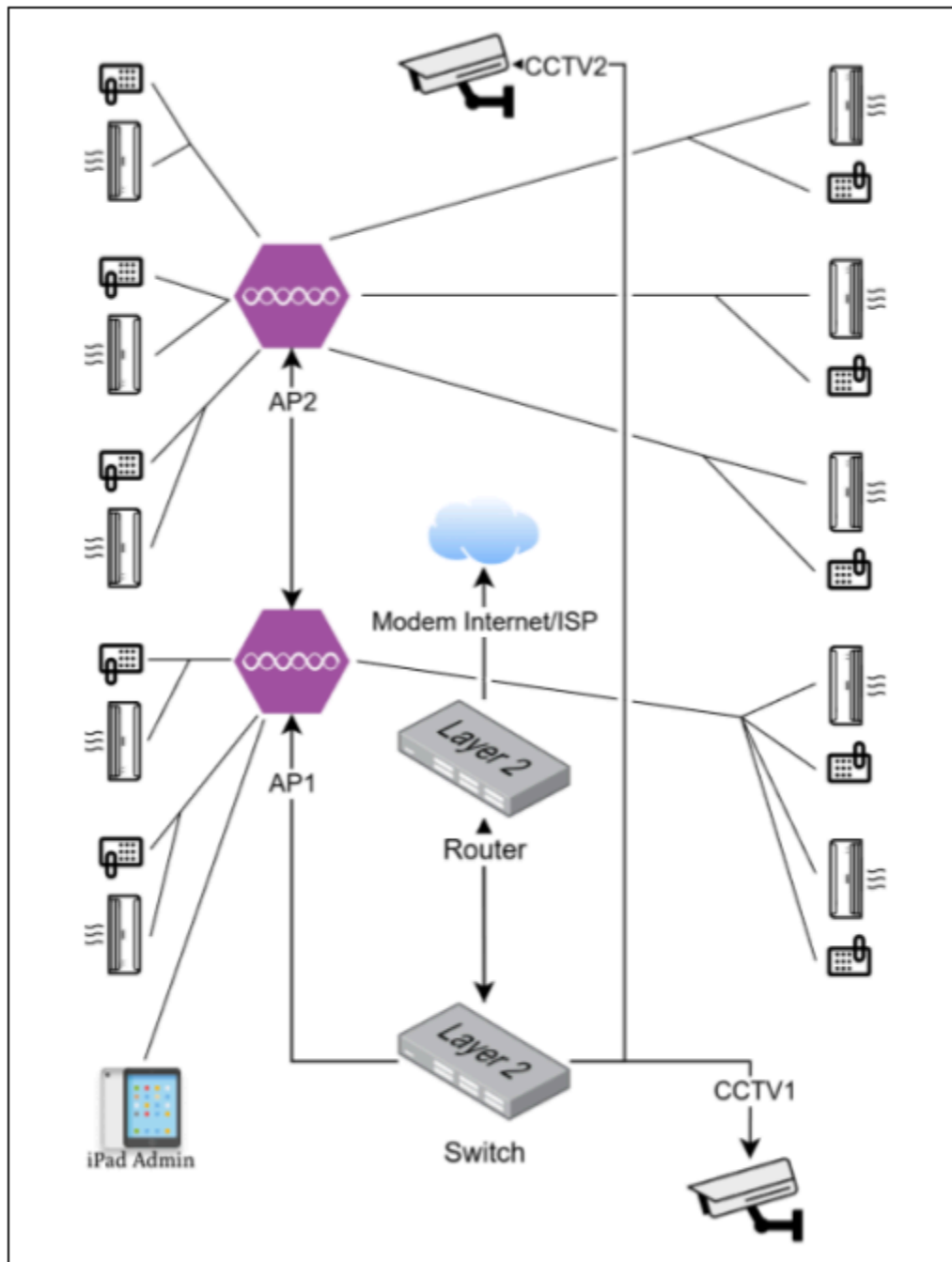
Number	Picture	Device Name	Qty	Hardware Specification
1		Mikrotik RB750Gr3(hEX)	2	Wireless Speed: 300 Mbps (2.4 GHz) Ethernet Ports: 1 × WAN, 4 × LAN (10/100/1000 Mbps) Max Power Consumption: 10W
2		TP-LINK TL-SG1016PE V5	1	Ethernet Ports: 16 × Gigabit RJ45 (10/100/1000 Mbps) PoE+ Ports: 8 × IEEE 802.3af/at, up to 30 W per port Total PoE Power Budget: 150W Features: PoE Auto Recovery, Plug-and-Play, Web UI & Management Utility Switching Capacity: 32 Gbps
3		Ruijie RG-RAP2260(G) Reye Wi-Fi 6	2	Wireless Speed: up to 1775 Mbps (574 Mbps @2.4 GHz + 1201 Mbps @5 GHz) Ethernet Ports: 2 × Gigabit (10/100/1000 Mbps) Max Power Consumption: ≤ 15.3 W Max Clients: up to 512 (recommended ~100 for best performance)
4		iPad	1	Chipset / SoC: Apple A16 Bionic Display / Layar: 11-inch Liquid Retina (2360 × 1640) Signal / Connectivity: Wi-Fi 6, Bluetooth 5.3, USB-C

5		Ezviz CCTV Camera Indoor 4MP C6N	2	Image Sensor: 5MP CMOS Maximum Resolution: 2560 × 1944 pixels
6		Polytron AC SMART Neuva Pro SMART AC 1/2 PK PAC 05VZSI	10	Cooling Capacity: ~ 5.000 BTU/h (0,5 PK) Power Consumption: ~ 390 Watt Electric Capacity / Voltage: 160-240 V, 50 Hz Refrigerant: R32
7		BARDI Smart Door Lock Z01	10	Unlock method: Fingerprint, PIN, RFID card, App, Mechanical key Power: 4× AA batteries (6V) Door thickness: 35–80 mm Capacity: Up to 100 users IP rating: IP20 (indoor use)

IP ADDRESS BOARDING HOUSE

No.	Device Name	Quantity	IP Address Range	Connection Type
1	Router	1	192.168.1.1	Gateway
2	Switch	1	-	LAN Distributor
3	Access Point	2	192.168.1.3 - 192.168.1.4	Wireless
4	iPad	1	192.168.1.10	Client Device (WiFi)
5	CCTV	2	192.168.1.20 - 192.168.1.21	LAN
6	Smart AC	10	192.168.1.30 - 192.168.1.39	Client Device (WiFi)
7	Smart Door Lock	10	192.168.1.40 - 192.168.1.49	Client Device (WiFi)

NETWORK TOPOLOGY DESIGN



BUDGET

DEVICES USED IN KOST					
No.	HARDWARE	PRODUCT	PRICE	Qty	TOTAL
1.	Switch	TP-LINK TL-SG1016PE V5	Rp 2.150.000	1	Rp 2.150.000
2.	Router	mikroTik RB750Gr3 (hEX)	Rp 950.000	1	Rp 950.000
3.	Access Point	Ruijie RG-RAP2260(G) Reyee Wi-Fi 6	Rp 1.700.000	2	Rp 3.400.000
4.	Tablet	11-inch iPad (A16) iBox Official Store iBox Online Store	Rp 6.699.000	1	Rp 6.699.000
5.	CCTV	Ezviz CCTV Camera Indoor 4MP C6N	Rp 450.000	2	Rp 900.000
6.	AC	Polytron AC SMART Neuva Pro SMART AC 1/2 PK	Rp 2.949.000	10	Rp 29.490.000
7.	Smart Door Lock	BARDI Smart Door Lock Z01	Rp. 929.000	10	Rp 9.290.000
The Total Price of The Devices			Rp 52.879.000		